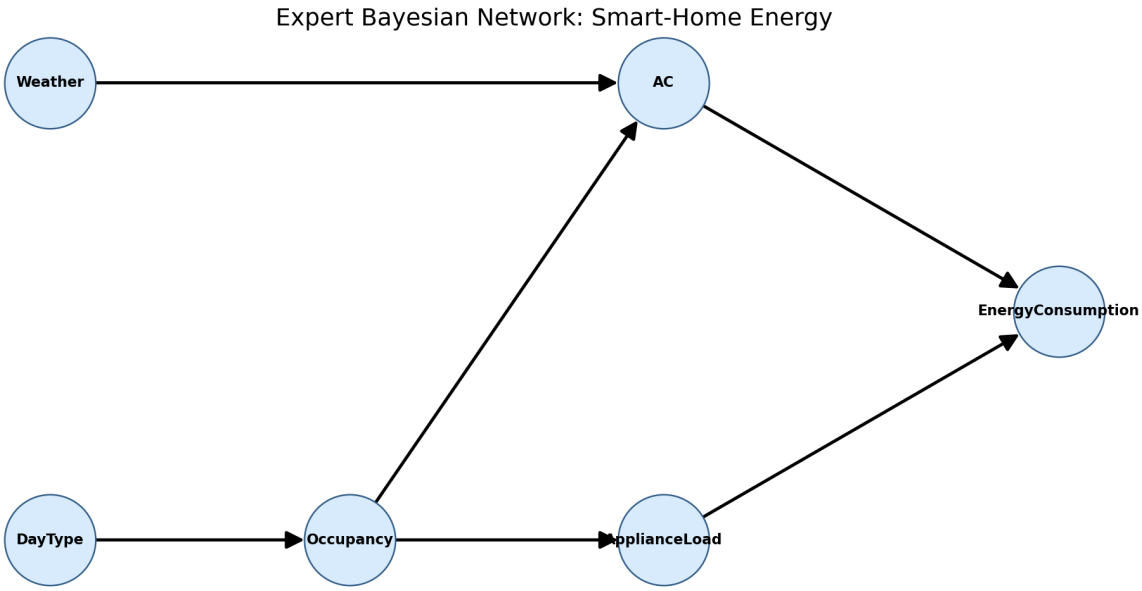
AI Group Assignment – 2

Bayesian Networks · Mars Rover PDDL · Resolution Refutation

Group members: [Add names and roll numbers]

1. Bayesian Network Analysis

Exact inference with a six-variable network, independently implemented CPT fitting and ancestral sampling with seed 2026. All expert CPTs are normalized.



Expert inference queries

Query	Probability
Q1	0.580631038
Q2	0.706265750
Q3	0.640198500
Q4	0.484250000
Q5	0.608188078
Q6	0.692866681
Q7	0.649899303

Q6 = 0.692866681; Q7 = 0.649899303. Conditioning on high energy opens the collider AC → Energy ← ApplianceLoad. Observing heavy load explains away some evidence for AC usage.

Hand calculation of Q1

P(O=High) = 0.70(0.45)+0.30(0.80) = 0.555. Sum over weather and occupancy of P(w)P(o) times the four AC/load combinations. The individual contributions are:

```
Hot High : P(o)=0.555000; P(E=High|w,o)=0.81825000; contribution=0.227064375
Hot Low : P(o)=0.445000; P(E=High|w,o)=0.56660000; contribution=0.126068500
```

```
Mild High : P(o)=0.555000; P(E=High|w,o)=0.60950000; contribution=0.101481750
Mild Low : P(o)=0.445000; P(E=High|w,o)=0.34452500; contribution=0.045994087
Cold High : P(o)=0.555000; P(E=High|w,o)=0.48425000; contribution=0.053751750
Cold Low : P(o)=0.445000; P(E=High|w,o)=0.29517500; contribution=0.026270575
TOTAL Q1 = 0.5806310375
```

Conditional independence

W independent of D: TRUE, paths blocked by A and E colliders.

W independent of D given E: FALSE, conditioning on the collider E activates a path.

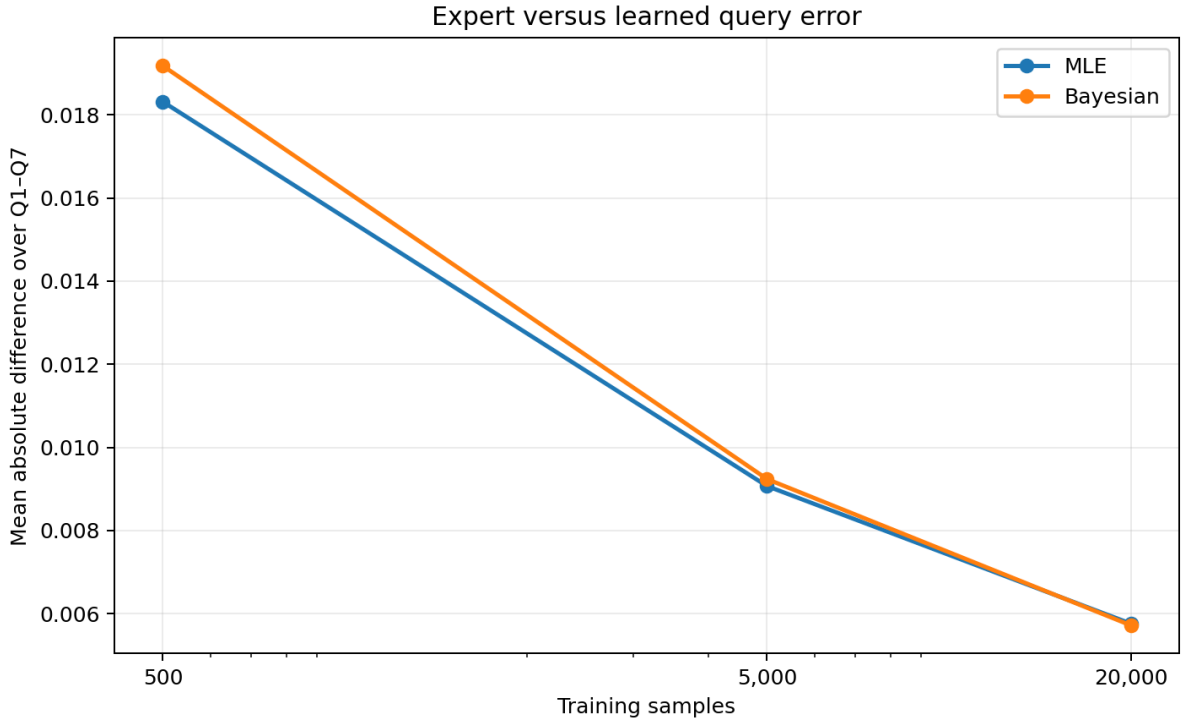
W independent of L given O: TRUE, paths are blocked at A/E or at conditioned non-collider O.

```
1 Weather independent DayType: 0.5000000000 vs 0.5000000000; equal=True
2 Weather independent DayType GIVEN E: 0.6081880785 vs 0.5997514833; equal=False
3 Weather independent Load GIVEN Occupancy: 0.5000000000 vs 0.5000000000; equal=True
```

Sampling & Parameter Learning

Sample sizes: 500, 5,000, 20,000. MLE estimates observed proportions; BDeu uses ESS = 10 divided uniformly across parent configurations/outcomes. Full datasets, CPT tables and query comparison CSVs are included.

n	W	frequency	O=High count	O=Low count
500	Hot	0.5040	140	112
500	Mild	0.3240	85	77
500	Cold	0.1720	43	43
5000	Hot	0.4940	1375	1095
5000	Mild	0.2976	786	702
5000	Cold	0.2084	580	462
20000	Hot	0.5053	5572	4534
20000	Mild	0.2972	3333	2610
20000	Cold	0.1976	2187	1764



n	Estimator	Mean |error|
500	MLE	0.0183164
500	Bayesian	0.0191810
5000	MLE	0.0090692
5000	Bayesian	0.0092379
20000	MLE	0.0057607
20000	Bayesian	0.0057195

AC CPT and inference comparison (n=500)

Weather/Occupancy	Expert	MLE	Bayesian BDeu
Hot/High	0.9000	0.8786	0.8741
Hot/Low	0.6000	0.5804	0.5792
Mild/High	0.4000	0.4353	0.4365
Mild/Low	0.1500	0.2468	0.2521
Cold/High	0.1000	0.1163	0.1306
Cold/Low	0.0500	0.0233	0.0410

Query	Expert	MLE	Bayesian	|Diff| MLE	|Diff| Bayesian
Q1	0.5806	0.5593	0.5582	0.0214	0.0225
Q2	0.7063	0.6595	0.6553	0.0468	0.0510
Q3	0.6402	0.6182	0.6130	0.0220	0.0272
Q4	0.4842	0.4745	0.4790	0.0097	0.0052
Q5	0.6082	0.5943	0.5877	0.0139	0.0205
Q6	0.6929	0.6987	0.6949	0.0058	0.0021
Q7	0.6499	0.6585	0.6557	0.0086	0.0058

AC CPT and inference comparison (n=5000)

Weather/Occupancy	Expert	MLE	Bayesian BDeu
Hot/High	0.9000	0.8880	0.8875
Hot/Low	0.6000	0.5881	0.5880
Mild/High	0.4000	0.4122	0.4124
Mild/Low	0.1500	0.1396	0.1405
Cold/High	0.1000	0.1017	0.1029
Cold/Low	0.0500	0.0455	0.0471

Query	Expert	MLE	Bayesian	|Diff| MLE	|Diff| Bayesian
Q1	0.5806	0.5840	0.5838	0.0034	0.0032
Q2	0.7063	0.7063	0.7058	0.0001	0.0005
Q3	0.6402	0.6442	0.6436	0.0040	0.0034
Q4	0.4842	0.5013	0.5016	0.0170	0.0173
Q5	0.6082	0.5975	0.5968	0.0107	0.0114
Q6	0.6929	0.6768	0.6765	0.0160	0.0164
Q7	0.6499	0.6376	0.6374	0.0123	0.0125

AC CPT and inference comparison (n=20000)

Weather/Occupancy	Expert	MLE	Bayesian BDeu
Hot/High	0.9000	0.9009	0.9008
Hot/Low	0.6000	0.5933	0.5933
Mild/High	0.4000	0.3936	0.3937
Mild/Low	0.1500	0.1567	0.1569
Cold/High	0.1000	0.0956	0.0959
Cold/Low	0.0500	0.0544	0.0548

Query	Expert	MLE	Bayesian	|Diff| MLE	|Diff| Bayesian
Q1	0.5806	0.5881	0.5881	0.0075	0.0075
Q2	0.7063	0.7087	0.7085	0.0024	0.0023
Q3	0.6402	0.6482	0.6480	0.0080	0.0078
Q4	0.4842	0.4967	0.4968	0.0125	0.0126
Q5	0.6082	0.6089	0.6087	0.0007	0.0005
Q6	0.6929	0.6875	0.6874	0.0054	0.0054
Q7	0.6499	0.6460	0.6459	0.0039	0.0040

At n=500 the largest MLE absolute error is Q2: 0.046787. Rare conditioned evidence and limited parent-configuration counts amplify sampling error.

At n=500 the largest Bayesian absolute error is Q2: 0.051008. Rare conditioned evidence and limited parent-configuration counts amplify sampling error.

BDeu smoothing can reduce variance in small samples, but may introduce bias; the measured errors above determine which estimator won for this random seed. Both generally improve as n grows.

2. Mars Rover PDDL Planning

Domain.pddl, Problem.pddl and Problem_extended.pddl use only :strips and :typing. Greedy best-first and optimal A* search implement the PDDL actions with capacity and directed connectivity, followed by full action replay validation.

Scenario	Search	Actions	Expanded	Time (s)	Goal
original	greedy	29	35	0.004006	PASS
original	optimal	25	1520	0.020247	PASS
extended	greedy	44	117	0.001268	PASS
extended	optimal	40	13156	0.33402	PASS

A* uses an admissible lower bound from the relaxed shortest route through outstanding collection locations plus unavoidable collect, deposit, and drill-pickup actions. Capacity restrictions are ignored in the heuristic, not in actual transitions.

The original optimal plan collects Rock2 and Rock4 together, then picks up the drill at SiteE and collects hard rocks Rock5/Rock3. The extended optimal plan reaches SiteB via Base → SiteA → SiteC → SiteD → SiteB and explicitly uses the one-way SiteC → SiteD slope; the Base ↔ SiteB route is closed.

Greedy searches fewer states but yields 29 vs 25 actions in the original scenario, and 44 vs 40 in the extended scenario. All final states have all rocks at Base, both bays empty and the rover at Base. Measured local Python planning times are reported; these are not external planner benchmark times.

Original optimal plan (25 actions)

```
(move Curiosity Base SiteB)
(collect-soft Curiosity Rock2 SiteB Bay1)
(move Curiosity SiteB SiteD)
(collect-soft Curiosity Rock4 SiteD Bay2)
(move Curiosity SiteD SiteB)
(move Curiosity SiteB Base)
(deposit Curiosity Rock2 Base Bay1)
(deposit Curiosity Rock4 Base Bay2)
(move Curiosity Base SiteA)
(move Curiosity SiteA SiteC)
(move Curiosity SiteC SiteE)
(pick-drill Curiosity Drill1 SiteE)
(move Curiosity SiteE SiteF)
(collect-hard Curiosity Rock5 SiteF Bay1)
(move Curiosity SiteF SiteE)
(move Curiosity SiteE SiteC)
(collect-hard Curiosity Rock3 SiteC Bay2)
(move Curiosity SiteC SiteA)
(move Curiosity SiteA Base)
(deposit Curiosity Rock3 Base Bay2)
(move Curiosity Base SiteA)
(collect-soft Curiosity Rock1 SiteA Bay2)
(move Curiosity SiteA Base)
(deposit Curiosity Rock1 Base Bay2)
(deposit Curiosity Rock5 Base Bay1)
```

Extended optimal plan (40 actions)

```
(move Curiosity Base SiteA)
```

```
(move Curiosity SiteA SiteC)
(move Curiosity SiteC SiteD)
(move Curiosity SiteD SiteB)
(collect-soft Curiosity Rock2 SiteB Bay1)
(move Curiosity SiteB SiteD)
(collect-soft Curiosity Rock4 SiteD Bay2)
(move Curiosity SiteD SiteE)
(pick-drill Curiosity Drill1 SiteE)
(move Curiosity SiteE SiteC)
(move Curiosity SiteC SiteA)
(move Curiosity SiteA Base)
(deposit Curiosity Rock2 Base Bay1)
(deposit Curiosity Rock4 Base Bay2)
(move Curiosity Base SiteA)
(move Curiosity SiteA SiteC)
(move Curiosity SiteC SiteE)
(move Curiosity SiteE SiteF)
(collect-hard Curiosity Rock5 SiteF Bay1)
(collect-hard Curiosity Rock7 SiteF Bay2)
(move Curiosity SiteF SiteE)
(move Curiosity SiteE SiteC)
(move Curiosity SiteC SiteA)
(move Curiosity SiteA Base)
(deposit Curiosity Rock5 Base Bay1)
(deposit Curiosity Rock7 Base Bay2)
(move Curiosity Base SiteA)
(move Curiosity SiteA SiteC)
(move Curiosity SiteC SiteE)
(collect-soft Curiosity Rock6 SiteE Bay1)
(move Curiosity SiteE SiteC)
(collect-hard Curiosity Rock3 SiteC Bay2)
(move Curiosity SiteC SiteA)
(move Curiosity SiteA Base)
(deposit Curiosity Rock3 Base Bay2)
(move Curiosity Base SiteA)
(collect-soft Curiosity Rock1 SiteA Bay2)
(move Curiosity SiteA Base)
(deposit Curiosity Rock1 Base Bay2)
(deposit Curiosity Rock6 Base Bay1)
```

3. Royal Archive Resolution Refutation

Goal: prove trusted(mira). Add negated goal not trusted(mira); eliminate implications and convert universally quantified formulas to CNF. Variables are standardized apart; no Skolemization is needed because there are no existential quantifiers.

```
ROYAL ARCHIVE - VERIFIED RESOLUTION REFUTATION
Negated goal: ~trusted(mira)
r1 from c1 + c10 => [~archivist(mira), ~verifies(mira,scroll_7)]
r2 from r1 + c9 => [~verifies(mira,scroll_7)]
r3 from r2 + c3 => [~finds_mark(mira,scroll_7), ~inspects(mira,scroll_7)]
r4 from r3 + c7 => [~finds_mark(mira,scroll_7)]
r5 from r4 + c8 => []
EMPTY CLAUSE [] DERIVED.
CONCLUSION: trusted(mira) is entailed.
```

The empty clause [] confirms a contradiction from the negated goal, so trusted(mira) follows. The Prolog source implements first-order resolution with unification; a separate Python checker was executed here. Native Prolog execution was not possible without an installed interpreter.

How to run

```
python Problem1_Bayesian/bayesian_energy.py
python Problem2_PDDL/rover_planner.py
python Problem3_Resolution/verify_resolution.py
swipl -q -s Problem3_Resolution/royal_archive_access.pl -g main -t halt
```